

Unit – 5

Singular Value Decomposition

Questions and answers

1. If $Q(x) = x^2 - 8xy - 5y^2$ then compute $Q(-3,1)$

Sol: $A = \begin{pmatrix} 1 & -4 \\ -4 & -5 \end{pmatrix}$ = symmetric matrix

$$Q(-3,1) = 28$$

2. Find the singular values of $A = \begin{pmatrix} 9 & -9 \\ -9 & 9 \end{pmatrix}$

Sol: Eigen values of $A = 0, 18$

$$\text{Singular values} = \sqrt{0}, \sqrt{18} = 0, 3\sqrt{2}$$

3. Test for positive definiteness $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$

Sol: Eigen values = $2, 2-\sqrt{2}, 2+\sqrt{2}$

$A=LDU$ decomposition is

$$U = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 4/3 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & -2/3 & 1 \end{pmatrix}$$

Pivots of $A = 2, 3/2, 4/3$

So, A is positive semi definite

4. Find the SVD of $A = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$

Sol: $AA^T = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

Eigen values = 1, 3

Singular values = 1, $\sqrt{3}$

Eigen vectors = $(1/\sqrt{2}) \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = U$

$$\varepsilon = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$V^T = (1/\sqrt{6}) \begin{pmatrix} 1 & -2 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Factorization of $A = U \Sigma V^T$ where,

$U = m \times m$ orthogonal matrix

$V = n \times n$ orthogonal matrix

$\Sigma =$ diagonal matrix

$$\Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}$$